

Mailing station

How do you send e-mail without a PC? It's simple. You take a certain device, plug it in any telephone jack and send and retrieve e-mail from your account. This device, for



which you have to pay Rs 5,990, is called an iStation and is manufactured by iNabling Technologies from Bangalore. If you want to print an e-mail, plug iStation in a printer and fire the print; You can even send mail in your native language. The iStation weighs 770 grams and comes equipped with a 256 KB capacity, an in-built modem, and a backlit LCD.

It's available in two models: Premier P1001 (Rs 5,990) and Classic C1001 (Rs 7,990). The Classic model has a printer and a PC port. iStation permits up to five users to use a single unit, each having their own private e-mail IDs.

3D MEMORY



A Silicon Valley start-up will try to popularise one of the more intriguing ideas of our time: three-dimensional semiconductors.

California-based Matrix has received strategic funding from Sony, Eastman Kodak and Microsoft, to release memory chips that contain many more layers of circuitry than ordinary chips. As a result, Matrix' chips resemble microscopic cubes rather than two-dimensional planes.

The chips will apparently incorporate some design techniques taken from digital flat panels. Matrix has borrowed many of its concepts for building vertical chips from the flat-panel display industry, which has perfected the art of mass-producing layered silicon films. The chips will be built using many

more layers of circuitry than current chips and as a result, resemble microscopic cubes instead of the existing chips which are much flatter.

The chips will be substantially smaller than competing products. Matrix claims that their technique will allow them to produce 64 MB flash memory cards that would sell for around \$10 (current 64 MB cards cost \$38).

It seems likely we will see these chips in devices relatively soon, considering the fact that Toshiba, Sony and Eastman Kodak are among the investors.

The transition to three dimensional chips is driven primarily out of a fear that Moore's Law will hit a wall. Designers will have to come up with other ways to allow chips to continue to become more powerful.

'I'm in a hurry'

You're a goner if your PC's managed to get this one. Goner is a powerful new Windows computer virus busy wreaking havoc and affecting over 17 countries, out of which the US, the UK and France have been the worst affected. The e-mail is named 'Hi' with a message that reads, "How are you? When I saw this screensaver I immediately thought of you. I'm in a hurry (sic). I promise you'll love it." It has an attachment called 'Gone.scr', a screensaver which displays a small dialog box which fools the user into believing that a program is running, while in the background, the system files are being deleted. The e-mail sends itself to the addresses in the Outlook Address Book. Goner can even disable anti-virus software in a computer. Watch out.

Patch work

Microsoft has released a security patch for Internet Explorer 5.5 and 6.0 to plug a hole that could allow hackers to steal passwords and trick people into downloading virulent files. The patch eliminates a wide range of previously known security problems and plugs three new holes. One problem affecting only IE 6.0, allows an attacker to alter HTML information in a way as to trick IE to open a damaging executable file without asking the person for confirmation. Another bug allows the hacker to gather personal information from the local system. A hacker could read, but not change, any file on the computer user's system that can be opened in a browser window.

Cool for laptops

Micro-fans developed at Purdue University, Indiana, could help in developing smaller and faster laptops and let manufacturers stash in more chips in devices than are currently possible. These fans, say the engineers, imitate the motion of a Chinese fan. These micro-

fans have polyester or even metal blades, which are "attached to tiny patches of piezoelectric ceramic". When an alternating current is applied, the ceramic expands and contracts and the blades move back and forth like a Chinese fan. However, there is scepticism that the airflow achieved by a ceramic fan will not cool

a chip satisfactorily. But they could be put to work in combination with rotational fans especially in spaces that are not reached by the latter. Engineers also claim that the piezoelectric fans consume far less power as compared to a rotational fan. Interestingly, being noiseless, they can also be used in cell phones.